

10.4 Tangent Planes & Linearization

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$$y - y_0 = \underbrace{\frac{dy}{dx}(x_0)}_{\text{slope}} (x - x_0)$$

$$z - z_0 = \underbrace{\frac{\partial f}{\partial x}(x_0, y_0)}_{\text{slope}} (x - x_0) + \underbrace{\frac{\partial f}{\partial y}(x_0, y_0)}_{\text{slope}} (y - y_0)$$

$$L(x, y) = \underbrace{\quad}_{\text{slope}} \underbrace{\quad}_{\text{slope}} \underbrace{\quad}_{\text{slope}} \underbrace{\quad}_{\text{slope}} + \underbrace{f(x_0, y_0)}_{z_0}$$

10.4.1

$$4) f(x, y) = \sin x + \cos y \quad (0, 0, 1) \quad f(0, 0) = \sin 0 + \cos 0 = 1$$

$$\frac{\partial f}{\partial x} = \cos x$$

$$\frac{\partial f}{\partial y} = -\sin y$$

$$\frac{\partial f}{\partial x}(0, 0) = \cos 0 = 1$$

$$\frac{\partial f}{\partial y}(0, 0) = \sin 0 = 0$$

$$z - z_0 = \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0)$$

$$z - 1 = 1(x - 0) + 0(y - 0)$$

$$10) f(x, y) = \ln(x^2 + y^2) \quad (1, 1, \ln 2)$$

$$\frac{\partial f}{\partial x} = \frac{2x}{x^2 + y^2}$$

$$\left. \frac{\partial f}{\partial x} \right|_{(1,1)} = \frac{2(1)}{(1)^2 + (1)^2} = 1$$

$$\frac{\partial f}{\partial y} = \frac{2y}{x^2 + y^2}$$

$$\left. \frac{\partial f}{\partial y} \right|_{(1,1)} = \frac{2(1)}{(1)^2 + (1)^2} = 1$$

$$z - \ln 2 = 1(x - 1) + 1(y - 1)$$

$$ax + by + cz + d = 0$$

$$27. f(x, y) = \ln(x^2 - 3y) \quad (1, 0) \quad f(1.1, 0.1)$$

$$\frac{\partial f}{\partial x} = \frac{2x}{x^2 - 3y} \quad \left. \frac{\partial f}{\partial x} \right|_{(1, 0)} = \frac{2(1)}{(1)^2 - 3(0)} = 2$$

$$\frac{\partial f}{\partial y} = \frac{-3}{x^2 - 3y} \quad \left. \frac{\partial f}{\partial y} \right|_{(1, 0)} = \frac{-3}{(1)^2 - 3(0)} = -3$$

$$f(1, 0) = \ln((1)^2 - 3(0)) = \ln(1) = 0$$

$$L(x, y) = 2(x - 1) + (-3)(y - 0) + 0$$

$$L(1.1, 0.1) = 2(1.1 - 1) + (-3)(0.1 - 0) + 0$$
$$= 0.2 - 0.3 + 0$$

$$= -0.1 \swarrow$$

$$f(1.1, 0.1) \approx -0.0943...$$

10.4.2 Jacobi Matrices

$\vec{f}(x, y) = \begin{bmatrix} f_1(x, y) \\ f_2(x, y) \end{bmatrix}$
denotes vector

$$J\vec{f}(x, y) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} \begin{matrix} \leftarrow f_1 \\ \leftarrow f_2 \end{matrix}$$

40. $\vec{f}(x, y) = \begin{bmatrix} e^x \sin y \\ e^{-y} \cos x \end{bmatrix} = \begin{matrix} f_1(x, y) \\ f_2(x, y) \end{matrix}$

$$\vec{f}(\vec{x}), \quad \vec{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$J\vec{f}(x, y) = \begin{bmatrix} e^x \sin y & e^x \cos y \\ -e^{-y} \sin x & -e^{-y} \cos x \end{bmatrix}$$

n inputs $\vec{x} \in \mathbb{R}^n$
 m outputs $\vec{f}(\vec{x}) \in \mathbb{R}^m$

$$J\vec{f}(\vec{x}) \in \mathbb{R}^{m \times n}$$

$m \times n$ matrix

$\uparrow$ rows $\uparrow$ columns

$$\underbrace{\vec{L}(\vec{x})}_{m \times 1} = \underbrace{J\vec{f}(\vec{x}_0)}_{m \times n} (\underbrace{\vec{x} - \vec{x}_0}_{n \times 1 - n \times 1}) + \underbrace{\vec{f}(\vec{x}_0)}_{m \times 1}$$

$$m \times \underbrace{n \quad n}_{\text{matrix multiplication}} \times 1$$

$$m \times 1 = m \times 1 + m \times 1$$

$$J\vec{f}(x,y) = \begin{bmatrix} e^x \sin y & e^x \cos y \\ -e^{-y} \sin x & -e^{-y} \cos x \end{bmatrix} \quad (0,0)$$

$$J\vec{f}(0,0) = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \quad \vec{f}(0,0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$L(\vec{x}) = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \underbrace{\left(\begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right)}_{\begin{bmatrix} x-0 \\ y-0 \end{bmatrix}} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0(x-0) + 1(y-0) + 0 \\ 0(x-0) + (-1)(y-0) + 1 \end{bmatrix}$$

10.5 Multivariate Chain Rule

$$w = f(x, y) = f(x(t), y(t))$$

$$x(t)$$

$$y(t)$$

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

$$1. \quad f(x, y) = x^2 + y^2 \quad x(t) = 3t \quad y(t) = e^t$$

$$\frac{\partial f}{\partial x} = 2x \quad \frac{\partial f}{\partial y} = 2y$$

$$\frac{dx}{dt} = 3 \quad \frac{dy}{dt} = e^t$$

$$\frac{dw}{dt} = 2x \cdot 3 + 2y e^t$$

$$= 2 \cdot 3t \cdot 3 + 2e^t \cdot e^t$$

$$t = \ln 2$$